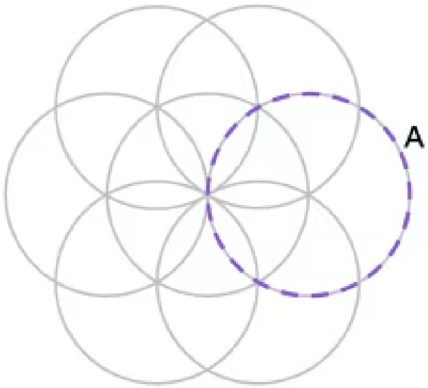
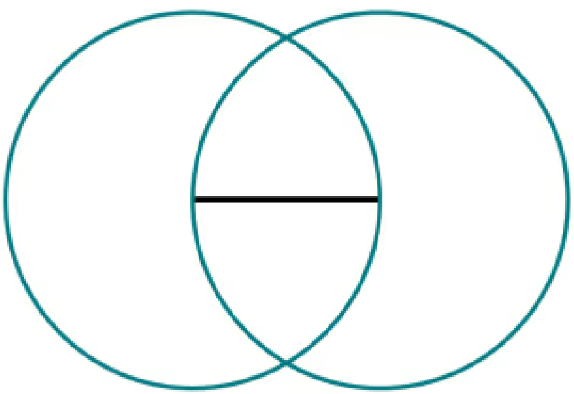
Problem solving with constructions

1 In this six-petal flower construction, at how many points does circle A intersect with at least one other circle?



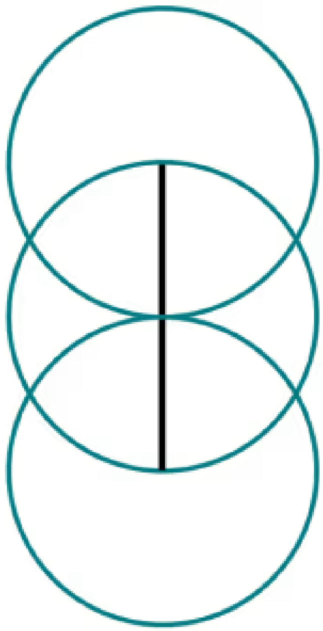
5

2 Which of these polygons can be formed by joining up combinations of intersections between any line segment or circle? (Tick **3** correct answers)



- [x] equilateral triangle
- [ ] trapezium
- [x] scalene triangle
- [x] rhombus
- [ ] square

3 Which of these polygons can be formed by joining up combinations of intersections between any line segment or circle? (Tick **4** correct answers)

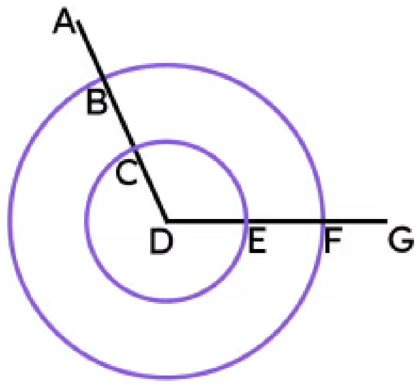


- [x] kite
- [x] parallelogram
- [ ] square
- [x] trapezium
- [x] pentagon

4 Which of these statements is true about the construction of a perpendicular bisector? (Tick **3** correct answers)

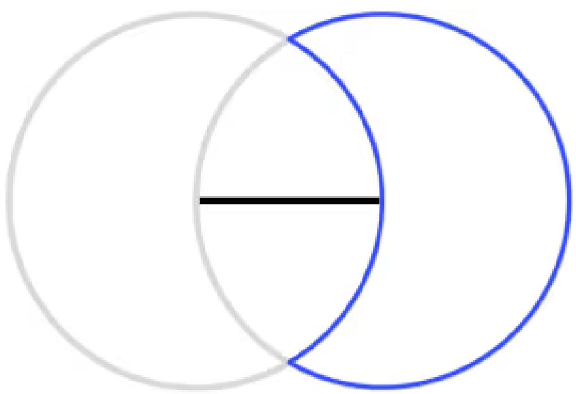
- [x] It is formed from the intersections of two congruent circles.
- [x] It is formed from intersections of different pairs of congruent circles.
- [ ] It can be formed from the intersections of any two circles.
- [ ] At least one circle must have its centre at the midpoint of a line segment.
- [x] All circles must have their centres at the endpoints of a line segment.

5 Which two pairs of points need to be joined with a line segment to find the fourth point in the kite? (Tick **2** correct answers)



- [ ] AE
- [x] BE
- [ ] CG
- [x] CF
- [ ] BC

6 This "crescent moon" logo can be formed using the construction of an equilateral triangle that uses two congruent circles of radius 9 cm. What is the perimeter of this logo (in cm, rounded to 1 dp)?



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